

Final Examination Certificate Master of Science



Mr. Amrina

born on April 01, 1994 in Darussalam

has successfully completed the Master exam
at the Saarland University, Faculty of Human and Business Sciences

The final examination has been passed: August 07, 2025.

Mr. Amrina has completed the

Master study course Educational Technology

with:

overall grade

VERY GOOD (1,4)

Master thesis entitled

Developing a Web-Based AI Tutor:
Integrating Generative AI and Retrieval-
Augmented Generation (RAG) for
Enhanced Learning Support

Master thesis was graded with

1,7

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Chairman Examination Committee

Univ.-Prof. Dr. phil. Armin Weinberger

Developing a Web-Based AI Tutor: Integrating Generative AI and Retrieval-Augmented Generation (RAG) for Enhanced Learning Support

Developer: Amrina

Context: Master's Thesis Project in Educational Technology (Saarland University), Germany.

Overview & Empirical Findings

ERM Tutor AI is an interactive, domain-specific conversational agent designed to support students in the Empirical Research Methods (ERM) course. Built in 2025, a period when generative AI models were not yet as highly advanced or inherently reliable as they are today, the system was engineered to act as a rigorously controlled, personalized academic mentor. It bridges the gap between static Learning Management Systems and students' need for on-demand, reliable, and conversational academic support.

Key Thesis Findings: Empirical evaluation of this system demonstrated its significant pedagogical value. In an evaluation conducted by a panel of expert instructors using a 7-point rubric, the RAG-based ERM Tutor AI statistically outperformed a baseline GPT-4o model in instructional accuracy ($M=6.75$ vs. $M=6.00$). A two-way ANOVA confirmed this difference as highly significant ($p < 0.001$). Qualitatively, the architecture successfully minimized AI hallucinations, ensuring that students received highly contextual, transparent, and curriculum-aligned guidance with superior information efficiency.

Core Problem & Motivation

Students tackling complex topics like research methodology often struggle with nuanced questions outside of class hours. General Large Language Models (LLMs) can easily hallucinate or provide information disconnected from the specific curriculum. ERM Tutor AI solves this by anchoring LLM capabilities directly to verified course materials, providing pedagogical scaffolding rather than just direct answers.

Technical Architecture

The application is deployed via Streamlit Cloud and relies on a robust backend stack:

- **Vector Retrieval (ChromaDB):** Course materials (PDFs, slides) are processed, chunked, and stored locally. The system utilizes Maximal Marginal Relevance (MMR) search to aggressively prioritize highly relevant local academic texts before referencing external knowledge.
- **Orchestration (LangChain):** Manages the flow between the user query, vector database, and the LLM.
- **Generative Model (OpenAI GPT-4o):** Powers the natural language understanding and generation, carefully guided by strict prompt engineering.

Key Pedagogical Features

1. **Strict Topic Adherence (Distraction-Free Learning):** A primary advantage of this application is its ability to keep students completely focused on the ERM curriculum. If a student asks a question unrelated to the course, the system explicitly refuses to answer. This strict boundary eliminates distractions and maintains a purely academic learning environment.
2. **Socratic Mentorship Persona:** The agent is engineered to respond not just as a search engine, but as an empathetic tutor. It provides factual answers, reflective insights, and engaging follow-up questions to stimulate critical thinking.
3. **Transparent Sourcing:** To build trust and encourage independent study, the UI dynamically extracts and displays the exact source document (e.g., specific PDF chapters) used to generate the answer.

Future Research Potential

This prototype serves as a strong foundation for broader doctoral research in AI-driven instructional design. Potential research directions include:

- **Evaluating Student Self-Efficacy:** Analyzing how continuous access to a localized RAG AI tutor impacts students' confidence and academic performance over a full semester.
- **Adaptive Scaffolding:** Expanding the RAG architecture to dynamically adjust its pedagogical tone and complexity based on a student's interaction history and knowledge gaps.
- **Trust & Hallucination Mitigation:** Investigating advanced retrieval techniques to completely eliminate AI hallucinations in strict academic settings.

Technical Stack: Python, LangChain, Streamlit, ChromaDB, OpenAI API.